

DORRA BEN AISSA

Agricultural Engineer – Plant Protection & Sustainable Farming Systems

dorrabenaissa4@gmail.com | +33 6 02 38 92 77 | Paris (open to national & international mobility)
French (C2) | English (C1 – TOEIC 860) | Arabic (native) | Driving licence (B) | Certiphyto



Profile — Agricultural engineer (AgroParisTech / INAT) specialised in plant protection and environment, with a strong commitment to agroecological transition and sustainable production systems. Multiple experiences in the design and monitoring of field trials, database building, advanced statistical analysis and pest monitoring, as well as project management and coordination, the design of decision-support tools and data valorisation. Committed to putting technology at the service of agriculture. Autonomous, rigorous and driven by turning knowledge into practical solutions for farmers.

PROFESSIONAL EXPERIENCE

Research & Experimentation Engineer, INRAE Grand-Est Colmar

Mar–Aug 2025 (6 months)

GrapeBreed4IPM programme, UMR SVQV, INRAE Grand-Est, Colmar

- Developed a **machine-learning image-analysis model** (ilastik + ImageJ) for the automated quantification of disease symptoms
- Developed an **R Shiny application** for scoring support, adopted by the research team
- Designed and monitored field trials: phenotyping of 254 genotypes, regular disease assessments, full data traceability
- Analysed and valorised data in R (ANOVA, linear models, QTL mapping) and reported results

Engineering Project – Project design & decision-support tool (PARSADA plan)

Sep 2024–Feb 2025 (6 months)

AgroParisTech, SupAgro Montpellier, AgroCampus Ouest

- Designed a **digital decision-support tool** for the integrated management of a pest: decision tree, intervention thresholds, recommendations tailored to field stakeholders
- Project management and set-up:** letter of intent, response to a call for projects, team coordination, reporting (poster, oral presentation)
- Identification of biocontrol and agroecological protection levers in a context of reduced synthetic inputs

Intern – Data Analysis & Food Systems

May–Jul 2024 (3 months)

SOUVNUTRI project – UMR PNCA INRAE, AgroParisTech

Built and managed an agricultural database, ensuring data structure and traceability | data analysis in R and Excel | analysis, summarisation and reporting of results to support decision-making

Intern – Entomology & Biological Control in Fruit Growing

May–Jun 2023 (2 months)

Citrus Technical Centre – Tunisia

Biological control: pest identification and monitoring, rearing and management of parasitoids/predators, implementation and evaluation of field releases

EDUCATION

Engineering Degree in Agronomy (Master's level) – AgroParisTech

2023–2025

Specialisation: Plant Protection and Environment (PPE) | Joint accreditation with SupAgro Montpellier, AgroCampus Ouest

Concentration: Production, Value Chains and Territories for Sustainable Development

Engineering Degree in Agronomy (Master's level) – National Agronomic Institute of Tunisia (INAT) 2021–2023

Specialisation: Plant Health and Crop Protection | Top of class | Double degree with AgroParisTech

SKILLS

- Data Analysis & Statistics:** Advanced R programming (ggplot2, Shiny, rqt), ANOVA, regression, linear models, QTL mapping, data visualisation, database management (MySQL), advanced Excel
- Experimental Design & Plant Trials:** protocol design, trial planning, multi-site experiments, field and greenhouse trials, phenotyping, scoring, data collection and traceability
- Crop Protection & Integrated Pest Management:** pest and disease monitoring, field diagnosis, biological control, biocontrol strategies, integrated pest management (IPM), Certiphyto certification
- Digital Tools & Decision Support:** development of R Shiny applications, design of decision-support tools, image analysis and automated phenotyping using ilastik and ImageJ
- Project Development & Coordination:** project planning, team coordination, preparation of project proposals, drafting technical deliverables, cross-functional collaboration and stakeholder engagement
- Scientific Communication:** technical reports, experimental results synthesis, scientific posters, oral presentations and technical communication materials
- Molecular Biology:** DNA/RNA extraction, PCR and qPCR
- Additional Tools:** QGIS, SPSS, Python (basic knowledge)

VOLUNTEERING & PROFESSIONAL INVOLVEMENT

- Secretary General – IEEE association (Tunisia):** promoting the use of technology in agriculture; participation in hackathons and in awareness and innovation initiatives
- Agricultural trade shows:** Paris International Agricultural Show (SIA) with Intercéréales and the French Beet Technical Institute; Tunisia Agricultural Show (SIAT) with the IEEE club

INTERESTS

Travel and cultural discovery | Regional-league handball (3 years) | Swimming